

Student Performance Analysis

Prepared by: Vishal Kumbhakar

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1. Project Overview & Objectives

Goal: To analyze the influence of socio-economic factors and educational background on student performance in Math, Reading, and Writing. **Objectives:**

- Identify the correlation between parental education and student grades.
 - Measure the effectiveness of the Test Preparation Course.
 - Detect performance gaps based on socio-economic indicators (Lunch programs).
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2. Dataset Description

- **File:** `students1.csv`
 - **Size:** 1,000 student records.
 - **Key Columns:** Gender, Race/Ethnicity, Parental Level of Education, Lunch, Test Preparation Course, Math Score, Reading Score, Writing Score.
 - **Calculated Fields:** Total Score, Average Score, Pass/Fail Status.
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3. Data Preparation & Cleaning (Python)

The following steps were performed in a Jupyter Notebook environment:

- **Metric Creation:** Generated a `Total_Score` (sum of all three subjects) and an `Average_Score`.

- **Logic Implementation:** Created a `Pass_Status` column where a score of **60 or above** is required to pass.
 - **Integrity Check:** Verified that all score columns were numeric and checked for missing values (none found).
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4. Key Exploratory Findings

- **The "Master's" Advantage:** Students whose parents hold a Master's degree achieved the highest average scores across all subjects.
 - **Preparation Impact:** Completing the Test Preparation Course resulted in a significant score boost, particularly in literacy-based subjects (Reading and Writing).
 - **Literacy Correlation:** There is an extremely high correlation (**0.95**) between Reading and Writing scores, indicating that these skills are developed concurrently.
 - **Gender Trends:** In this dataset, female students consistently outperformed male students in Reading and Writing, while Math scores were more competitive.
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5. Power BI Dashboard Design

The interactive dashboard was designed to provide a "drill-down" experience for school administrators.

Key DAX Measures Used:

- Average Overall Score = `AVERAGE('Table'[Average_Score])`
- Pass Rate % = `DIVIDE(CALCULATE(COUNTROWS('Table'), [Average_Score] >= 60), COUNTROWS('Table'))`

Visual Layout:

- **Top Row:** KPI Cards for Total Students, Avg Score, and Pass Rate.
- **Middle Left:** Bar Chart showing the linear relationship between Parental Education and Grades.
- **Middle Right:** Comparative Bar Chart showing the "Prep Course" bonus per subject.

- **Bottom:** Scatter Plot showing the distribution of Reading vs. Writing excellence.

6. Conclusion & Recommendations

- **Recommendation 1:** Subsidize or mandate the Test Preparation Course for at-risk groups, as it is the most effective way to raise scores.
- **Recommendation 2:** Focus on early math intervention for students from backgrounds where parental education stopped at High School.
- **Recommendation 3:** Use literacy-based tutoring to address writing gaps, as reading and writing skills are deeply linked.

7. Annex – Dashboard Screenshots

